

AWS Application Load Balancer (ALB)

What is a Load Balancer?

- An AWS Load Balancer automatically **distributes incoming web traffic** across multiple EC2 instances to ensure that no single server is overwhelmed.
- It improves the availability, scalability, and fault tolerance of your applications.
- Users access your website using the **DNS name** of the **Application Load Balancer (ALB)**.
- The **ALB receives the request** and checks which EC2 instance is **healthy and least loaded**.
- It then **routes the request** to that instance.

- If one EC2 instance fails, the **ALB automatically reroutes** traffic to healthy instances — ensuring **zero downtime** for users.

Step 1. Create 2 EC2 Instances

The screenshot shows the AWS Management Console 'Launch an instance' page. The 'Number of instances' field is set to 2. The 'Amazon Linux' AMI is selected under the 'Quick Start' tab. The 'Launch instance' button is highlighted.

Note:

- Add the port no 80 and 22 in security group and
- Also fill “User Data” field which is used to provide a **script that will run**

automatically when the instance is launched for the first time.

This process is called: **Bootstrapping**

Lastly Launch Instance >

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
# Use this for your user data (script from top to bottom)
# Install httpd (Linux 2 version)
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello World from ${hostname -f}</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-006b4a3ad5f56fbd6

Virtual server type (instance type)
t3.micro

Firewall (security group)
SSH and HTTP

Storage (volumes)
1 volume(s) - 8 GiB

[Free tier](#): In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't)

[Cancel](#) [Launch instance](#) [Preview code](#)

2 Instance got created, now rename second instance name **“My Second Instance”**.

Instances (1/2) [Info](#) Last updated 1 minute ago [Connect](#) [Instance state](#)

[All states](#)

	Name ↗	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm statu
☐	My First Instance	i-0cdd7a98a3e9e90df	Running 🔍 🔍	t3.micro	Initializing	View alarm
☒	My First Ins... ↗	i-0d938603a26bd2219	Running 🔍 🔍	t3.micro	Initializing	View alarm

Copy the Ip addresses of both instances and paste in 2 different browser tabs.

i-0cdd7a98a3e9e90df (My First Instance)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ **Instance summary** [Info](#)

Instance ID i-0cdd7a98a3e9e90df	Public IPv4 address 13.48.84.96 open address	Private IPv4 addresses 172.31.21.193
IPv6 address -	Instance state Running	Public DNS ec2-13-48-84-96.eu-north-1.compute-1.amazonaws.com open address
Hostname type	Private IP DNS name (IPv4 only)	

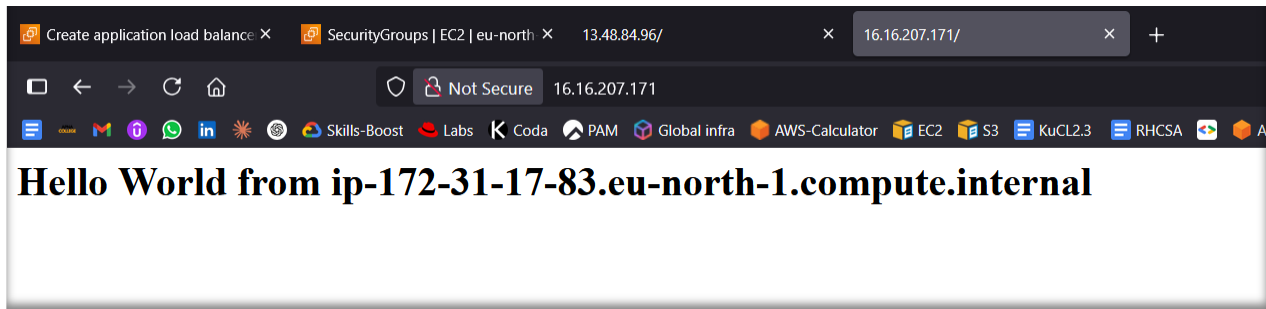
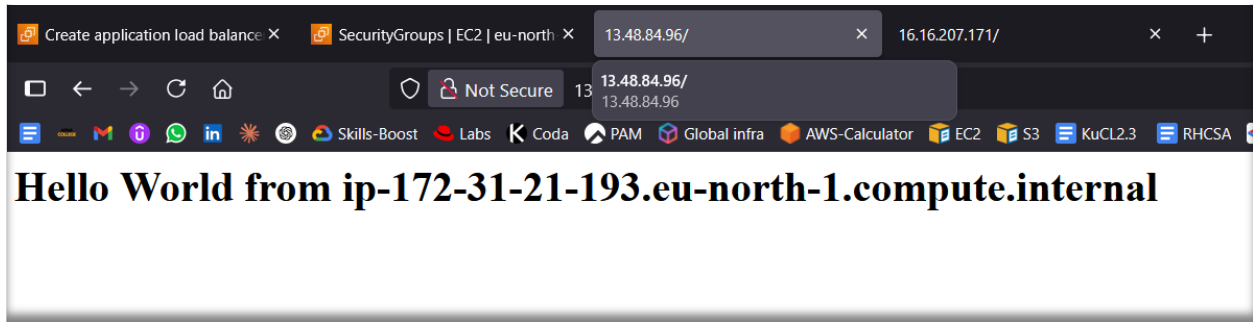
i-0d938603a26bd2219 (My First Instance)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ **Instance summary** [Info](#)

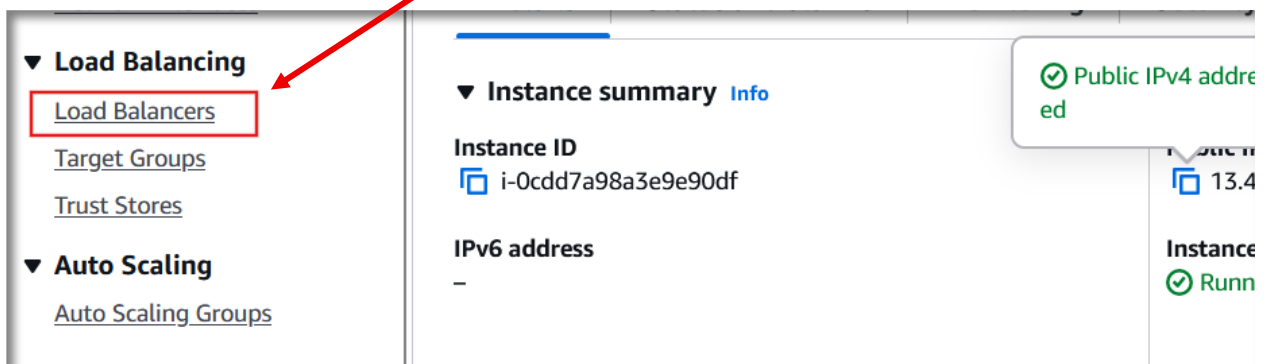
Instance ID i-0d938603a26bd2219	Public IPv4 address 16.16.207.171 open address	Private IPv4 addresses 172.31.17.83
IPv6 address -	Instance state Running	Public DNS ec2-16-16-207-171.eu-north-1.compute-1.amazonaws.com open address
Hostname type	Private IP DNS name (IPv4 only)	

- You can see the instance is on 2 different browser tabs.
- It do not act like Load balancing on single tab.



Here is the example of website hosting using AWS instance.

Step 2. Go to EC2 Load Balancing and create Load Balancer.



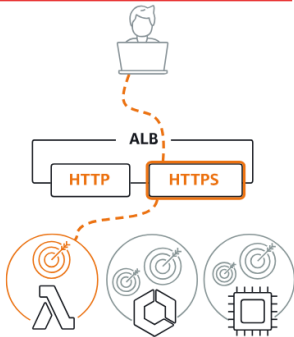
Create the Application Load Balancer

Compare and select load balancer type

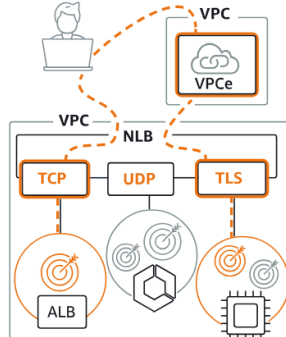
A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer [Info](#)



Network Load Balancer [Info](#)



Gateway Load Balancer [Info](#)



Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Inst-1-2-demo-load-balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Select All the availability zones.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to

☒ eu-north-1a (eun1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0564382b012af7878

IPv4 subnet CIDR: 172.31.16.0/20

☒ eu-north-1b (eun1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0dd8290178d80e454

IPv4 subnet CIDR: 172.31.32.0/20

☒ eu-north-1c (eun1-az3)

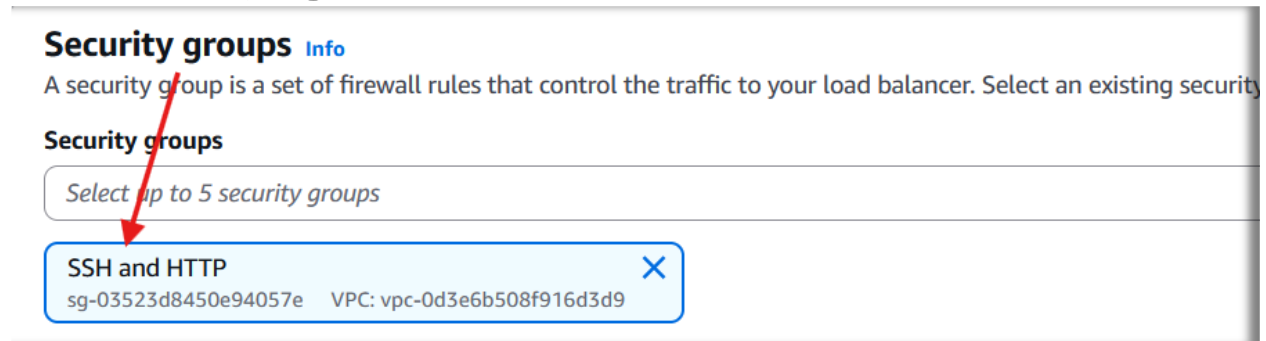
Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0ae3d0fb8e13d5a67

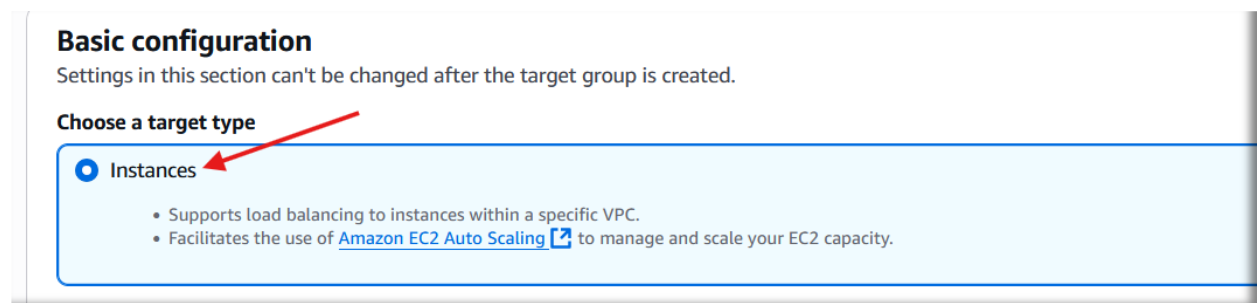
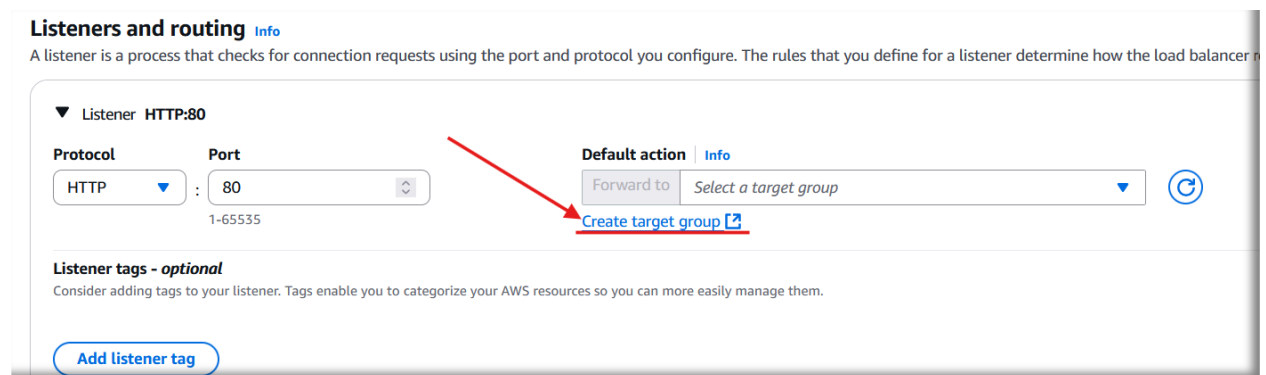
IPv4 subnet CIDR: 172.31.0.0/20

Enable the port number 80 and 22 in Security group.



Now Create the Target Group.

Target Group is a logical grouping of backend resources (like EC2 instances) that a Load Balancer sends traffic to.



Target group name

demo-target-grp

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol
Protocol for load balancer-to-target communication. Can't be modified after creation.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

to easily manage them.

Cancel Next

Select the instances to make target groups

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-0cdd7a98a3e9e90df	My First Instance	Running	SSH and HTTP	eu-north-1a
☒	i-0d938603a26bd2219	My Second Instance	Running	SSH and HTTP	eu-north-1a

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (2) Remove all pending

Show only pending < 1 > ⚙️

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-0cdd7a98a3e9e90df	My First Instance	80	Running	SSH and HTTP	eu-north-1a	172.31.21.193	subnet-0564382b012
i-0d938603a26bd2219	My Second Instance	80	Running	SSH and HTTP	eu-north-1a	172.31.17.83	subnet-0564382b012

2 pending Cancel Previous Create target group

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes traffic.

▼ Listener **HTTP:80**

Protocol HTTP **Port** 80 1-65535

Default action [Info](#)

Forward to Select a target group ↻

[Create target](#) HTTP

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can manage them more effectively.

Add listener tag

You can add up to 50 more tags.

Target group is created, now click on create the Load Balancer.

✓ **Successfully created load balancer: Inst-1-2-demo-load-balancer**

It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

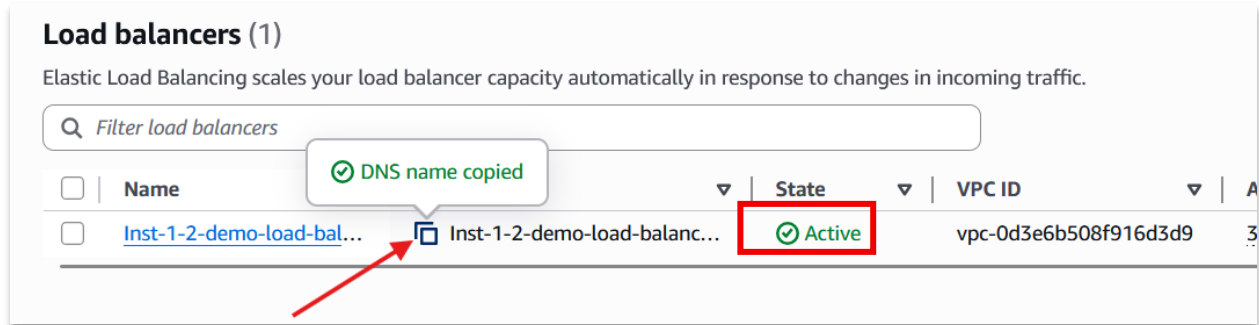
ℹ️ **Application Load Balancers now support public IPv4 IP Address Management (IPAM)**

You can get started with this feature by configuring IP pools in the [Network mapping](#) section. Edit IP pools

Inst-1-2-demo-load-balancer ⌂ Actions

▼ **Details**

Load balancer type Application	Status Provisioning	VPC vpc-0d3e6b508f916d3d9	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z23TAZ6LKFMNIO	Availability Zones subnet-0ae3d0fb8e13d5a67 eu-north-1c (eun1-az3) subnet-0564382b012af7878 eu-north-1a (eun1-az1)	Date created June 4, 2025, 09:50 (UTC+05:30)



✓ Conclusion:

Through this hands-on setup, I learned:

- 🔄 How traffic is intelligently distributed across multiple EC2 instances
- 📦 The importance of Target Groups and Health Checks for backend reliability
- ⚙️ Practical deployment flow involving EC2, Security Groups, ALB, and DNS
- 📊 Real-time performance benefits and automatic failover capability of ALB

Load Balancer is not just a tool—it's the backbone of reliability in large-scale web systems.